

Multivariable Calculus

By now, we should be familiar with derivatives and integrals of single-variable functions. ...

Further, I'm going to assume that we're on board with the idea of 3d+ spaces, functions, etc. These are notes for doing calculus in more than two dimensions, not notes about how higher-dimensional spaces work to begin with. So, we should already know how parametric equations and vectors are a useful tool when working in higher dimensions.

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1. Multivariable Limits

Just as we are introduced to single-variable calculus by beginning with limits, it is useful to see multivariable calculus first through the lens of multivariable limits. We won't end up *directly* needing most of this later, but an intuitive understanding of this behavior will help us understand the differences between multivariable and single-variable infinitesimals.

First, let's ask what it means to take the limit of some function $f(x, y)$. Well, recall that the limit of $g(x)$ as x goes to a is the value that g approaches as the distance between x and a decreases. Similarly, the limit of $f(x, y)$ as (x, y) goes to (a, b) is the value that f approaches as the distance between (x, y) and (a, b) decreases.

(TODO) visuals

This makes multivariable limits seem very similar to single-variable ones, and they are, but there's a catch. With a 1d limit, or the limit of a single-variable function, x can only approach a in two ways: from above or from below. With a 2d or higher limit, (x, y) can approach the point (a, b) from infinite different directions or along any possible path through a 2d plane. Now, recall that with 1d limits, both approaches (above and below) must agree in order for the limit to exist. Similarly, with multivariable limits, **every approach must agree in order for the limit to exist**. As you might imagine, this makes it more difficult to prove that limits exist.

(TODO) techniques for solving, factoring out, a var, grouping and substituting, parametrics, higher-dimensions, a generalized approach?, iterated limits, squeeze theorem techniques

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